Derivatives and Fixed Income
PGE M1 Fall 2026

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Session 10: Black-Scholes

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Exam

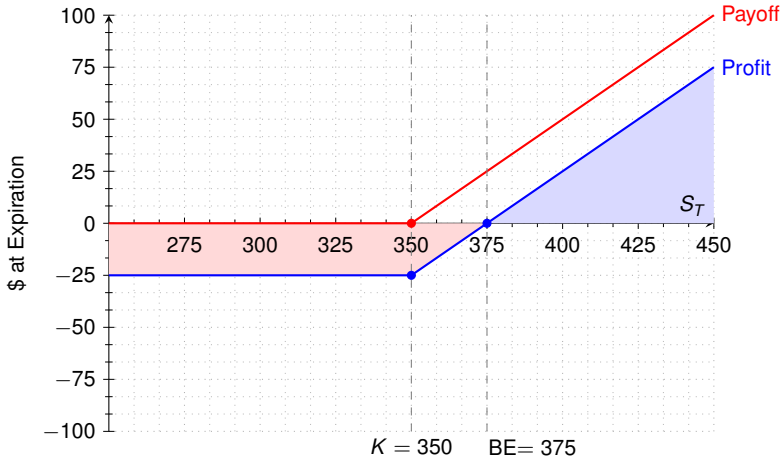


Midterm

- Covers material from sessions 1-8 + today (Black-Scholes)

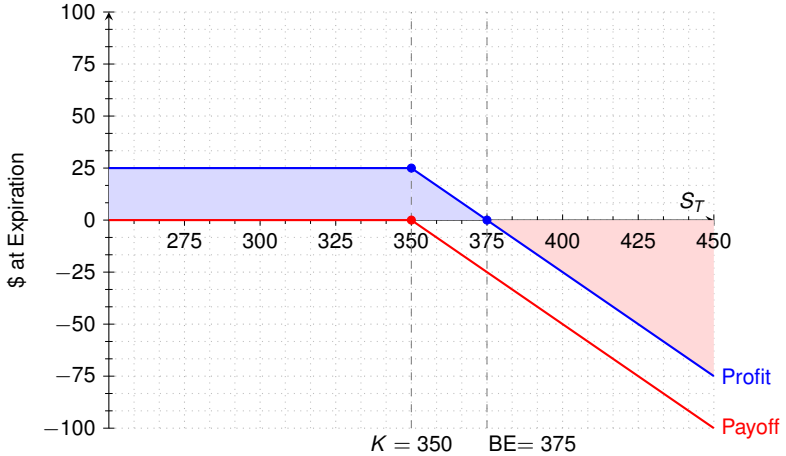
Long Call

Suppose you entered a long call position to buy Apple stock in one year at \$350, paying a premium of \$23.81. Assume $r = 5\%$, so $23.81 \times 1.05 = 25$. Draw the payoff and profit diagrams.



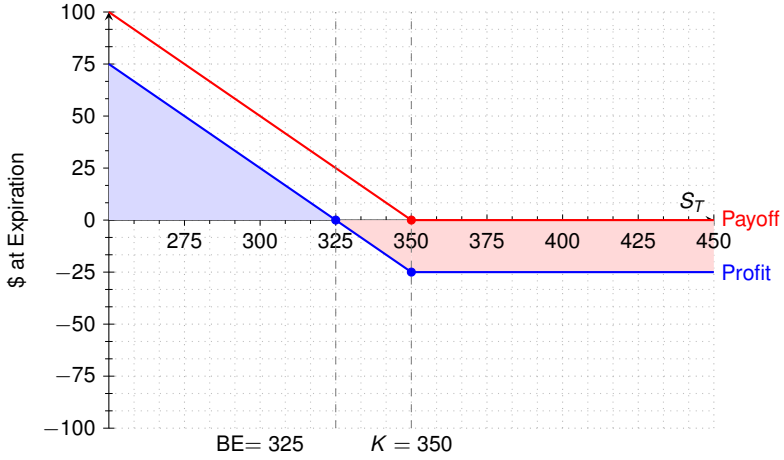
Short Call

Suppose you entered a short call position to sell Apple stock in one year at $350, receiving a premium of $23.81. Draw the payoff and profit diagrams.



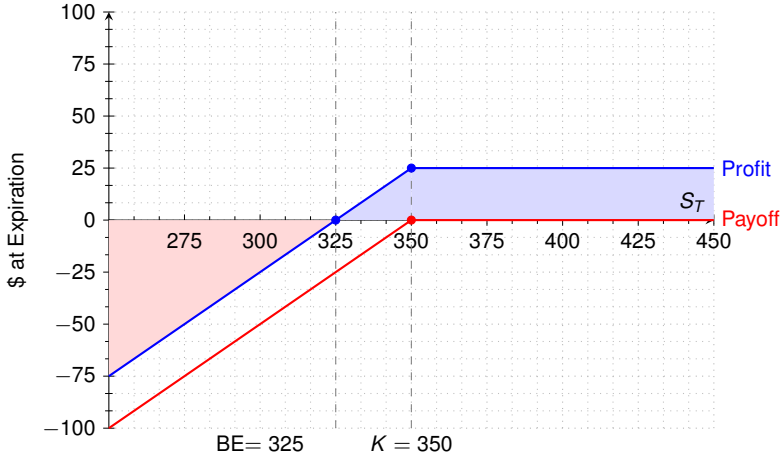
Long Put

Suppose you entered a long put position to sell Apple stock in one year at \$350, paying a premium of \$23.81. Assume $r = 5\%$, so $23.81 \times 1.05 = 25$. Draw the payoff and profit diagrams.



Short Put

Suppose you entered a short put position to buy Apple stock in one year at \$350, receiving a premium of \$23.81. Assume $r = 5\%$, so $23.81 \times 1.05 = 25$. Draw the payoff and profit diagrams.



P&L for Calls and Puts

	Max Profit	**Max Loss**	**Break-even**
Long Call	∞	$-c$	$K + c$
Short Call	$+c$	$-\infty$	$K + c$
Long Put	$K - p$	$-p$	$K - p$
Short Put	$+p$	$-(K - p)$	$K - p$

Expected Values (**)



Suppose you enter a bet. With probability 50%, you will receive €2 in one year, and 0 otherwise. What is the fair value of this bet if it were traded in financial markets? Assume $r = 5\%$.

$$\text{price of bet today} = \frac{\text{Expected Value in 1 year}}{1+r} = \frac{1}{1.05} = 0.95$$

The continuous discounting version would be

$$\text{price of bet today} = \text{Expected Value in 1 year} \times e^{-rT}$$

A European Call (**)

A European call option with expiration date T and strike price K has a payoff of $\max(S_T - K, 0)$. The call price is the expected value of this bet,

$$c = e^{-rT} \times \mathbb{E}^* \left[\max(S_T - K, 0)\right]$$

We can rewrite this using the indicator function $\mathbb{1}_{S_T > K}$ as

$$\begin{aligned} c &= e^{-rT}\mathbb{E}^*\left[(S_T - K)\mathbb{1}_{S_T>K}\right] \\ &= e^{-rT}\mathbb{E}^*\left[S_T\mathbb{1}_{S_T>K} - K\mathbb{1}_{S_T>K}\right] \\ &= e^{-rT}\mathbb{E}^*\left[S_T\mathbb{1}_{S_T>K}\right] - Ke^{-rT}\mathbb{E}^*\left[\mathbb{1}_{S_T>K}\right] \end{aligned}$$

A European Call (**)

From

$$c = e^{-rT}\mathbb{E}^*\left[S_T \mathbb{1}_{S_T>K}\right] - Ke^{-rT}\mathbb{E}^*\left[\mathbb{1}_{S_T>K}\right]$$

- $\mathbb{E}^*\left[\mathbb{1}_{S_T>K}\right]$ is simply the probability that $S_T > K$ at T, $\Pr(S_T > K)$
- From the definition of conditional expectation, $\mathbb{E}^*\left[S_T \mathbb{1}_{S_T>K}\right] = \mathbb{E}^*\left[S_T \middle| S_T > K\right] \times \Pr(S_T > K)$
- $$c = e^{-rT} \cdot \Pr(S_T > K) \cdot \left(\mathbb{E}^*\left[S_T \middle| S_T > K\right] - K\right)$$

A European Call (**)

$$c = e^{-rT} \cdot \Pr(S_T > K) \cdot \left(\mathbb{E}^* \left[S_T \middle| S_T > K \right] - K \right)$$

The value of the call is time-discounted expected value of the terminal stock price conditional on the call being ITM minus the strike, scaled by the probability of being ITM.

A European Call (**)

Defining

$$N(d_2) \doteq \Pr(S_T > K)$$

and

$$N(d_1) \doteq \frac{\mathbb{E}^*\left[S_T \middle| S_T > K\right]}{\mathbb{E}^*[S_T]} \times \Pr(S_T > K)$$

where $\mathbb{E}^*[S_T] = S_0 \times e^{rT}$,

$$c = S_0 N(d_1) - K e^{-rT} N(d_2)$$

Black-Scholes formula for the price of a European call

$$c = S_0 N(d_1) - K e^{-rT} N(d_2)$$

where $N(\cdot)$ is the normal CDF,

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

and S_0 is the spot price, r is the risk-free rate and σ is the volatility of stock prices, with the strike K and maturity T.

In the Apple Excel file.. (ignore dividends)

Scenario Inputs	
Strike price (K)	**$350.00**
Days to expiry (T, in days)	**182**

Market Inputs	
Underlying price (S)	**$340.00**
Risk-free rate (r)	**4.18%**
Dividend yield (q)	**0.32%**
Volatility (σ)	**24.69%**

Black-Scholes-Merton Formula Steps	
Time to maturity, T (years) = days / 365	0.4986
$d_1 = [\ln(S/K) + (r - q + \sigma^2/2) \cdot T]/(\sigma \cdot \sqrt{T})$	0.0313
$d_2 = d_1 - \sigma \cdot \sqrt{T}$	-0.1430
$N(d_1)$ — standard normal CDF	0.5125
$N(d_2)$ — standard normal CDF	0.4431

Option Prices	
Call price $= S \cdot e^{-qT} \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2)$	$22.07

Black-Scholes Model Assumptions



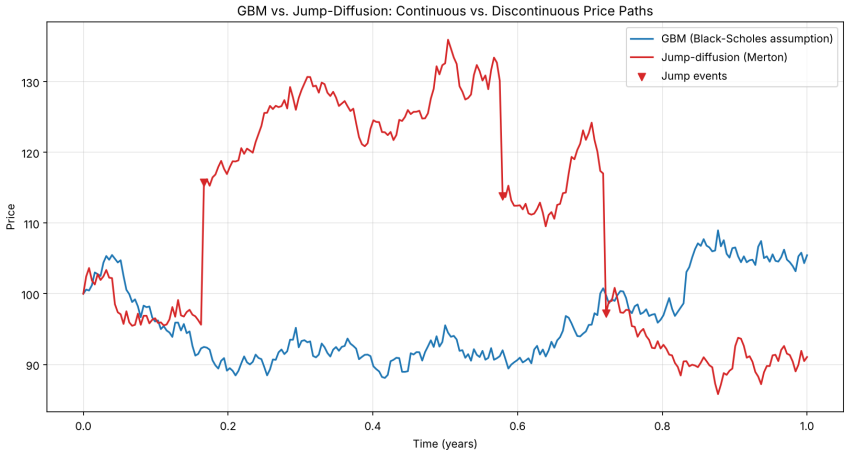
BS assumes

- Stock prices & returns are log-normally distributed
  - Prices follow a geometric Brownian motion
- Volatility of returns is constant
- There is continuous trading
- There are no transaction costs, no taxes
- There are no dividends paid on the underlying
- There are no riskless arbitrage opportunities
- Short-selling is permitted
- r is fixed
- For European options

Black-Scholes vs the Real World

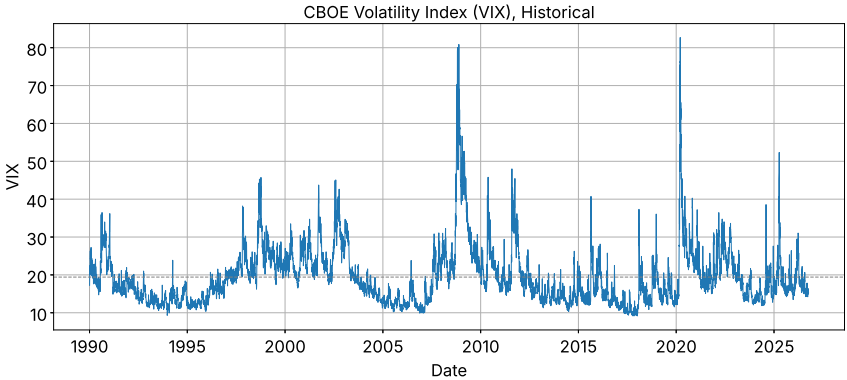


Black-Scholes vs the Real World (**)



Information about companies arrives discontinuously, so stock prices in reality are closer to a “jump-diffusion” process rather than a GBM.

Black-Scholes vs the Real World (**)



VIX is the annualized expected volatility of the S&P 500 index over the next calendar month. On average, it's $20 \times \sqrt{30/365} = 5.75\%$ per month.

Implied Volatility



In the Black-Scholes formula for the price of a European call,

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

S_0, r, σ, K, T are "inputs". Of these 5 inputs, σ is the only unobserved variable.

Based on the observed market value of the call (and put) option(s), we can infer the implied volatility of the underlying stock that produces the observed market price based on the BS formula. The implied vol then, is the forward-looking estimate of the stock's volatility.

Black-Scholes formula for the price of a European put



While the European call is given by

$$c = S_0 N(d_1) - K e^{-rT} N(d_2)$$

the European put is given by

$$p = K e^{-rT} N(-d_2) - S_0 N(-d_1)$$

where $N(-d_1) = 1 - N(d_1)$, or once c is known,

$$p + S_0 = K e^{-rT} + c$$

Exercise



Exercise 15.3 (**)

Calculate the price of a 3-month European put option on a non-dividend-paying stock with a strike price of \$50 when the current stock price is \$50, the risk-free interest rate is 10% per annum, and the volatility is 30% per annum.

$$p = Ke^{-rT}N(-d_2) - S_0N(-d_1)$$

Using $K = 50, S_0 = 50, \sigma = 30\%, r = 10\%, T = 0.25$,

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} = \frac{\ln(50/50) + (0.1 + 0.09/2)0.25}{0.3\sqrt{0.25}} = 0.2417$$

$$d_2 = d_1 - \sigma\sqrt{T} = d_1 - 0.3\sqrt{0.25} = 0.0917$$

Plugging back in,

$$p = 50e^{-.1(0.25)}N(-0.0917) - 50N(-0.2417) = 2.37$$

Exercise 15.3 using Apple Options Grid file (**)

Scenario Inputs	
Strike price (K)	**$50.00**
Days to expiry (T, in days)	**91**

Market Inputs (linked live from the top of this tab)	
Underlying price (S)	**$50.00**
Risk-free rate (r)	**10.00%**
Dividend yield (q)	**0.00%**
Volatility (σ)	**30.00%**

Black-Scholes-Merton Formula Steps	
Time to maturity, T (years) = days / 365	0.2493
$d_1 = [\ln(S/K) + (r - q + \sigma^2/2) \cdot T]/(\sigma \cdot \sqrt{T})$	0.2413
$d_2 = d_1 - \sigma \cdot \sqrt{T}$	0.0915
$N(-d_1)$	0.4046
$N(-d_2)$	0.4635

Option Prices	
Put price $= K \cdot e^{-rT} \cdot N(-d_2) - S \cdot e^{-qT} \cdot N(-d_1)$	**$2.37**

Properties of the BS value of European calls & puts



Holding r, σ, K, T fixed, let $S_0 \to \infty$.

The call will almost surely be exercised, while the put will not.

$N(d_1), N(d_2) \to 1$,

$$c = S_0 - Ke^{-rT}$$

and

$$p = 0$$

Properties of the BS value of European calls & puts



Holding S_0, r, K, T fixed, let $\sigma \to 0$.

There is no more uncertainty regarding the stock price.

If $S_0 > Ke^{-rT}$,

$$c = S_0 - Ke^{-rT}$$

while

$$p = 0$$

If $S_0 < Ke^{-rT}$,

$$c = 0$$

while

$$p = Ke^{-rT} - S_0$$

American vs European Options



For an American call option on a non-dividend paying stock, it is never optimal to exercise early.

Proof. Recall the lower bound on the value of a European call,

$$c \geq S_0 - Ke^{-rT}$$

Since an American option is at least as valuable as its European counterpart, $C \geq c$. Moreover, if the option were to be exercised today, the payoff would be

$$S_0 - K$$

Since $r > 0$,

$$C \geq c \geq S_0 - Ke^{-rT} > S_0 - K$$

So $C > S_0 - K$ when $T > 0$ (there is time remaining before expiry) and $C = c$.

American vs European Options



For an American call option on a dividend paying stock, it may be optimal to exercise early.

- You can re-invest the dividends depending on the timing.

For an American put option on a non-dividend paying stock, it may be optimal to exercise early.

- Since you sell at the strike price, you can re-invest the proceeds right away.

PCP with Dividends (again)

Recall PCP with the present value of dividends

$$p + S_0 = \frac{K}{(1+r)^T} + c + D$$

You can generally re-arrange,

$$p = \frac{K}{(1+r)^T} + c + D - S_0$$

and

$$c = p + S_0 - \frac{K}{(1+r)^T} - D$$